



Graph Coloring

PPT PRESENTED BY

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What is Graph Coloring?

Graph Coloring is the assignment of labels ('colors') to the vertices of a graph such that no two adjacent vertices (connected by an edge) receive the same color.

Formal Definition

Given graph $G = (V, E)$:

V = set of vertices

E = set of edges

A proper k -coloring is a function:

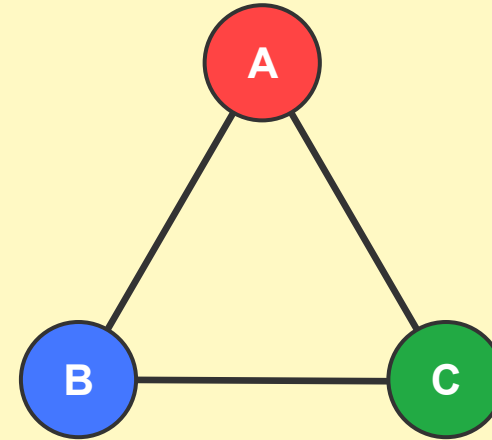
$c : V \rightarrow \{1, 2, \dots, k\}$

such that $c(u) \neq c(v)$ whenever $\{u, v\} \in E$

A graph is k -colorable if a proper k -coloring exists for it.

Note: Colors are just labels — the actual colors used don't matter, only that adjacent vertices differ.

Illustrative Example: K_3 (Triangle)



K_3 : 3 vertices, all connected

- A–B edge: $A \neq B$ ✓
- B–C edge: $B \neq C$ ✓
- A–C edge: $A \neq C$ ✓

Minimum colors needed = 3

(Cannot use only 2 colors!)



Chromatic Number $\chi(G)$

The chromatic number $\chi(G)$ of a graph G is the MINIMUM number of colors required for a proper vertex coloring.

Graph Type	Example	$\chi(G)$	Reason
Complete Graph K_n	K_4 (4 mutual edges)	n	Every vertex connects to all others; all need distinct colors
Bipartite Graph	Even cycle C_4	2	Vertices split into 2 independent sets; 2 colors suffice
Odd Cycle C_{2k+1}	Triangle C_3	3	Odd-length cycle cannot be 2-colored (parity argument)
Tree or Forest	Any acyclic graph	2	Trees are bipartite; BFS/DFS gives 2-coloring
Planar Graph	Any map graph	≤ 4	Four Color Theorem: every planar graph is 4-colorable

Greedy Coloring Algorithm

Greedy Coloring: Process vertices one by one and assign the SMALLEST available color not used by any already-colored neighbor.

Algorithm Steps

For each vertex v in order $v_1, v_2, \dots, v_n$:

1. Find all colors used by v 's neighbors that have already been colored.
2. Assign to v the smallest color NOT in that set.

Time Complexity:

$O(V + E)$ — linear in graph size

Limitation:

NOT always optimal — result depends on vertex ordering. May use MORE colors than the chromatic number.

Trace: Graph with 5 vertices

Vertex	Colored Neighbors	Assigned Color	Reason
v_1	—	Red (1)	No neighbors colored yet
v_2	v_1 (Red)	Blue (2)	Red is taken by v_1
v_3	v_1, v_2	Green (3)	Red+Blue taken
v_4	v_2, v_3	Red (1)	Blue+Green taken; Red free!
v_5	v_1, v_4	Blue (2)	Red taken by v_1, v_4

The Four Color Theorem

Four Color Theorem: Every planar graph can be properly colored using at most **4 colors**.

History of the Proof

1852: Francis Guthrie first conjectured the theorem while coloring a map of England.

1879: Kempe published a 'proof' — found to be flawed by Heawood in 1890.

1976: Kenneth Appel and Wolfgang Haken (Univ. of Illinois) produced the first correct proof — with a computer checking 1,936 special cases.

Significance: First major theorem PROVEN using a computer. 124 years from conjecture to proof!

Map Coloring Interpretation

Any geographic map can be colored with at most 4 colors so that no two adjacent regions (sharing a border) have the same color.

Key: A map corresponds to a PLANAR graph —

- Vertices = regions/countries
- Edges = shared borders
- Planar = can be drawn without edge crossings

Note: 5 Colors were known to suffice (1890). The hard part was proving 4 is enough!

5-Color Theorem (Easy Proof via Induction):

Every planar graph has a vertex of degree ≤ 5 (by Euler's formula). Remove this vertex, 5-color the rest by induction (IH), then reinsert the vertex — with at most 5 neighbors using ≤ 5 colors, at least one color remains available for the removed vertex.

Applications in Computer Science

Graph coloring models many real-world conflict-avoidance and resource-allocation problems:

Register Allocation in Compilers

Variables in a program form an 'interference graph' — two variables are adjacent if their lifetimes overlap (they cannot share a register). Coloring this graph assigns variables to CPU registers. Vertices needing more colors than available registers are 'spilled' to memory.

Frequency Assignment in Networks

In mobile networks, nearby cell towers must operate on different frequencies to avoid interference. Model towers as vertices and proximity as edges. Graph coloring assigns frequencies (colors) while minimizing the total frequency spectrum used — saving expensive bandwidth.

Scheduling with Conflicts

Exam scheduling: students (or courses) with common students are 'conflicting' — they cannot be in the same time slot. Build a conflict graph and color it. Each color represents a time slot. Minimum colors = minimum number of time slots needed.

Sudoku as Graph Coloring

A 9×9 Sudoku grid is a graph with 81 vertices (cells). Two vertices are adjacent if they share a row, column, or 3×3 box. We need to color this graph with 9 colors (digits 1-9) such that each row, column, and box uses all 9 colors exactly once.



THANK YOU

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